

ENGINEERING CASE STUDY: Open-Circuit Fault Simulation & Failsafe Analysis

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Platform: 2018 Ford F-150 2.7L EcoBoost	Fault Code Simulated: DTC P0113 (Intake Air Temperature Circuit High)
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SITUATION

The vehicle presented with an illuminated Malfunction Indicator Lamp (MIL) immediately upon key-on, exhibiting extended cranking time and a conservative default power delivery map, with the cooling fans commanding high-speed operation despite normal engine coolant temperatures. A preliminary diagnostic scan isolated an active DTC P0113 (Intake Air Temperature Sensor 1 Circuit High), and review of the Freeze Frame data confirmed the fault was logged at zero engine RPM (key-on, engine-off), indicating a static circuit failure rather than a dynamic performance drift.

TASK

The objective was to systematically isolate the root cause between a wiring harness short, a PCM failure, or a localized sensor defect, and to characterize how the PCM's failsafe logic compensated for the loss of its primary air density input within the speed-density calculation architecture.

ACTION

I configured a live FORScan telemetry session and captured three discrete datasets: a healthy baseline with the IAT/TIP sensor connected, an active-fault dataset with the sensor pigtail deliberately open-circuited at the charge pipe, and a verification dataset following circuit restoration and a Keep Alive Memory (KAM) reset. Short Term Fuel Trim (SHRTFT), secondary intake air temperature (IAT2), and calculated engine load were logged continuously across all three states to isolate the PCM's substitute-value behavior and resulting fuel compensation strategy.

RESULT

Failsafe System Evidence:

- Short Term Fuel Trim (SHRTFT1) Shift: -0.68% (Baseline) $\rightarrow +0.57\%$ (Active Fault) to prevent lean detonation.
- Multi-Sensor Isolation: Secondary manifold sensor (IAT2) remained active, tracking heat soak from 41°C to 48°C .
- Closed-Loop Restoration: SHRTFT returned to -2.2% immediately following Keep Alive Memory (KAM) reset.

F-150 2.7L EcoBoost: IAT/TIP Open-Circuit Failsafe Analysis

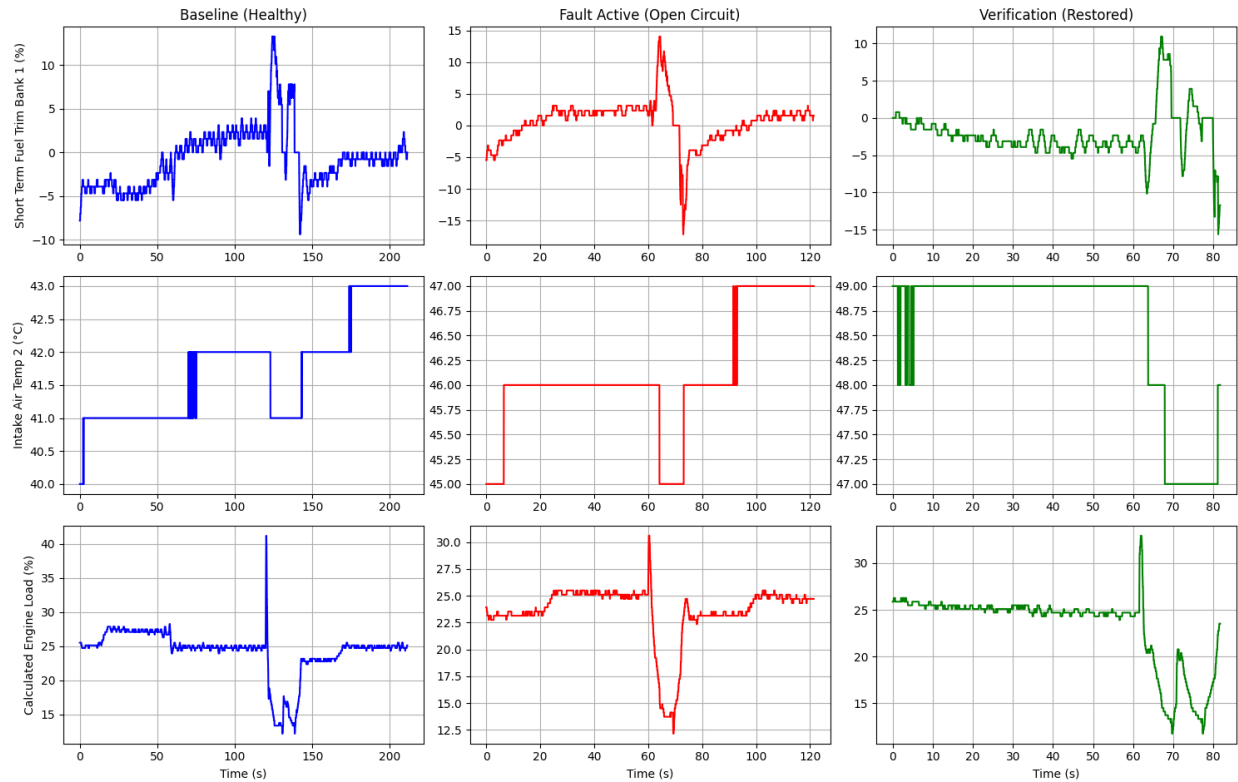


Figure 2.1: Side-by-side comparison of Baseline, Active Fault, and Verification telemetry.

SYSTEMS ENGINEERING & DIAGNOSTIC FINDINGS

- Closed-Loop Fuel Trim Compensation:** With the IAT/TIP circuit open, the PCM lost its primary air-density input and could no longer trust its speed-density fueling calculation. Rather than risk a lean-detonation event under boost, it abandoned tight stoichiometric targeting and biased SHRTFT1 from a -0.68% baseline trim to $+0.57\%$ under active fault, commanding a safely richer mixture. The sharpest transient enrichment (exceeding $+12\%$ momentarily) lines up with the high-load excursion captured mid-run, showing the compensation strategy scales with instantaneous load rather than applying a single fixed offset.
- Sensor Circuit Isolation:** Because IAT/TIP and IAT2 are physically separate sensors on distinct circuits, the secondary post-intercooler sensor (IAT2) remained fully active throughout the fault window, tracking real heat soak from 41°C at baseline to 48°C during the fault and verification runs. This redundancy was the key diagnostic clue: a live, trending IAT2 signal alongside a flat, defaulted IAT/TIP reading confirmed the failure was isolated to a single open circuit rather than a shared harness ground, a PCM-side fault, or a broader sensor-suite failure.
- Verification Protocol:** Following physical restoration of the sensor connector, a Keep Alive Memory (KAM) reset was performed, and the DTC transitioned from Active to Stored/Historic. A subsequent run cycle confirmed IAT/TIP telemetry resumed active ambient tracking and SHRTFT1 stabilized to -2.2% , closely matching the pre-fault baseline. The MIL remained

extinguished through the full verification cycle, confirming the repair fully restored closed-loop fueling control rather than merely clearing a stored code.